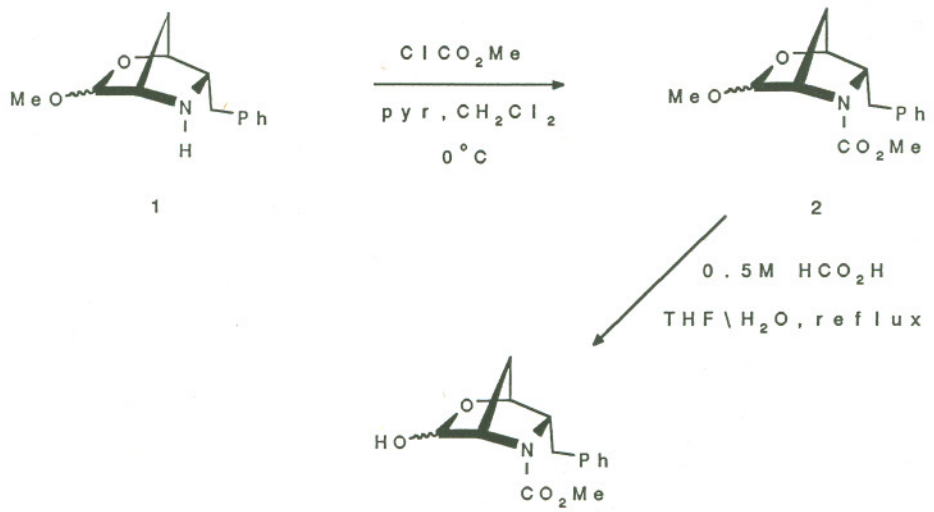
ORGANIC CHEMISTRY CUME

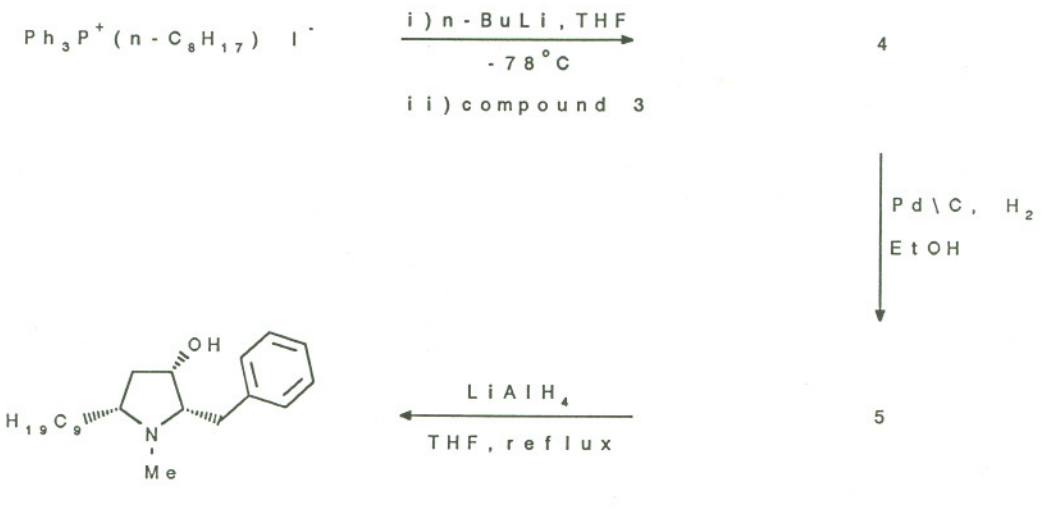
March 2nd 1991

All questions were taken from C.S. Pak et al J. Org. Chem. 1991, **56**, 1128.

1. Draw the mechanism of the following two step transformation. (30pts) Show all stages of the mechanism, including all proton transfers. Explain why the carbamate structure in compound **2** is not hydrolysed under these conditions. What conditions would be necessary to remove such a group? (10pts)



2. Compound **3** was transformed into the naturally occurring antifungal alkaloid (+)-Preussin via a three step sequence. The reagents have been given to you as well as the structure of (+)-Preussin. Predict the structure of species **4** and **5**. (20pts) Comment on the selectivity of the transformation **3** to **4**, (10pts) and draw the mechanism of the transformation of **5** to **6**. (10pts) Standard work-ups are assumed after each step.



3. A standard means of probing the interaction of an enzyme and an inhibitor of that enzyme is to test a series of structurally analogous species as inhibitors.If you knew that Preussin inhibited a particular enzyme E, and you were interested in optimizing this inhibition what simple structural changes would you suggest in order to probe this inhibition of enzyme E? Give two simple changes that you might try and explain why you chose those particular structures. (20pts)